GGG-A-145a
August 6, 1965

SUPERSEDING
Int. Fed. Spec. GGG-A-00145 (GSA-FSS)
February 4, 1964

FEDERAL SPECIFICATION

ADJUSTING TOOL, BRAKE SHOE (SPOON TYPE)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers tools for adjusting brake shoes on automotive vehicles (see 6.1). The adjusting tool consists of a steel bar with flat, angled blade ends for insertion through slots in brake back plates to adjust the brake shoes.

1.1.1 *Federal specification coverage.* Federal specifications do not include all varieties of the commodity as indicated by the title of the specification, or which are commercially available, but are intended to cover only those generally used by the Federal Government.

1.2 Classification.

1.2.1 *Type and sizes.* The brake shoe adjusting tool shall be of one type and the following sizes as specified (see 6.2):

Size 1—6- to 8-inch length.
Size 2—8-1/2- to 9-3/4-inch length.

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids, or request for proposal, form a part of this specification:

Federal Specification:

GGG-S-121—Screwdriver and Screw Starter, Hand.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.
Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).
Fed. Test Method Std. No. 151—Metals, Test Method.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-H-15424—Hand Tools, Packaging of.

Military Standards:

MIL-STD-105—Sampling Procedures

FSC 5120

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and Tables for Inspection by Attributes.

MIL-STD-129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Illustrations. The illustrations herein are descriptive and not restrictive and are not intended to preclude the purchase of brake tools otherwise conforming to this specification.

3.2 Material. The brake shoe tool shall be forged from high grade alloy steel free from imperfections that might affect serviceability. It shall be plated or otherwise treated to resist corrosion.

3.3 Hardness. The tool, including the angled and tapered ends, shall be properly heat treated to show a hardness value of not less than 42 nor more than 48 on the Rockwell C scale.

3.4 Bending moment. The tools shall withstand a minimum bending moment test of 500 inch-pounds when tested in accordance with 4.4.2.

3.5 Blade-end toughness. The blade ends shall comply with the toughness test specified in 4.4.3.

3.6 Design. The brake adjusting tool shall be a steel bar round, flat, or otherwise shaped for adjusting brakes on automotive vehicles. The tool shall be angled, curved, offset, shaped, and tapered on each end in a manner to permit brake adjustment without injury to the person using the tool (see fig. 1). At the option of the contractor, the tool may be notched, radiused, or slotted on the tapered end(s) of the tool to prevent interference while making brake adjustment. When specified (see 6.2), special configurations shall be provided such as one side notched or radiused, one end slanted, or other special features.

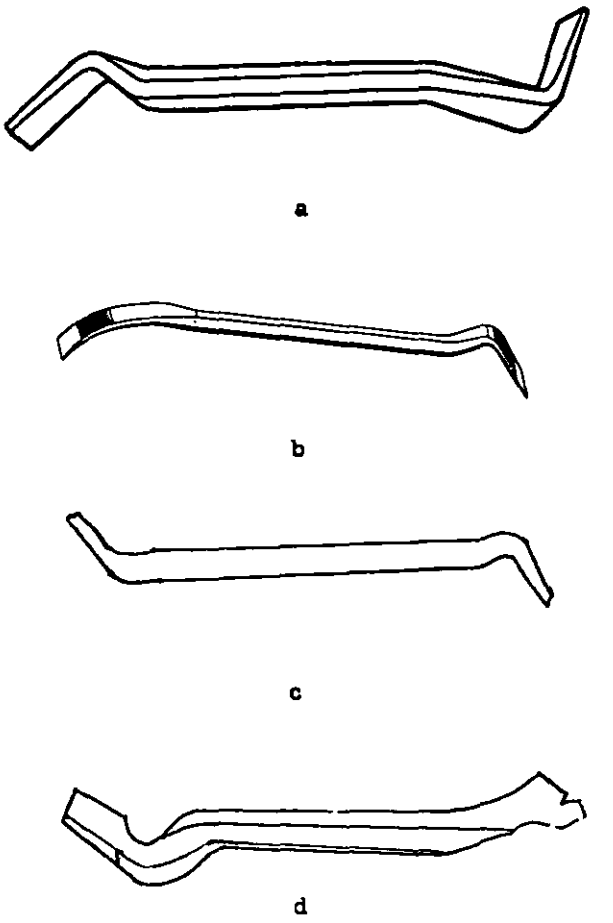


FIGURE 1. Brake adjusting tool. (Alternate designs.)

3.7 Size 1 brake tool. Size 1 brake adjusting tool shall be curved, offset, angled, or shaped and tapered in a manner to permit easy adjustment of star nuts on brakes with limited brake adjusting space (see 4.5.1).

3.7.1 *Length.* Size 1 brake tool shall be not less than 6 inches nor more than 8 inches in overall length.

3.7.2 *Body dimension.* Body dimension

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shall be a minimum of 7/16-inch diameter or 5/16-inch across flats.

3.8 Size 2 brake tool. Size 2 brake tool shall be designed for easy brake adjustment of star nuts on brakes where space is not limited. The spoons shall be angled, offset, and curved sufficiently to clear flanges and wheel diameters (see 4.5.2).

3.8.1 *Length.* Size 2 brake tool shall be not less than 8-1/2 inches nor more than 9-3/4 inches in overall length.

3.8.2 *Body dimension.* Body dimension shall be a minimum of 7/16-inch diameter, or 5/16-inch across flats.

3.9 Finish. The adjusting tool shall be free from visible defects such as scale, seams, laps, burrs, nodules, cracks, and other detrimental defects.

3.10 Coating. Tools shall be coated to prevent rust in accordance with the manufacturer's standard practice.

3.11 Identification marking. Each adjusting tool shall be marked in a plain and permanent manner with the manufacturer's name or with a trademark of such known character that the source of manufacture may be readily determined.

3.12 Workmanship. Workmanship shall be first class. Brake tools shall be smoothly finished, and free from any defects which may adversely affect their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 *Inspection of materials and components.* In accordance with 4.1, the supplier is responsible for insuring that materials and components used were manufactured, tested, and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified herein, or, if none, in accordance with this specification.

4.2 Sampling procedures. Sampling procedures shall be in accordance with MIL-STD-105. Data for sampling shall be as stated in table I.

4.3 Examination.

4.3.1 *Visual examination.* Each sample unit shall be examined for any nonconformance in design, material, finish, coating, workmanship, and marking.

4.3.2 *Dimensional examination.* Each sample unit shall be examined for any nonconformance with dimensional requirements.

TABLE I. *Sampling data*

Category	Sample unit	Inspection level	Acceptable quality level	AQL expressed in terms of	Reference
Visual examination	1 each	II	6.0	Defects per 100 units	4.3.1
Dimensional examination	1 each	S-4	1.5	Defects per 100 units	4.3.2
Testing	1 each	S-3	1.0	Percent defective	4.4 4.5
Preparation for delivery	One container	S-2	4.0	Defects per 100 units	4.6

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4.4 **Material testing.** Each sample unit shall be tested in accordance with 4.4.1, 4.4.2, and 4.4.3.

4.4.1 *Hardness.* Hardness test shall be conducted in accordance with method 243.1 of Fed. Test Method Std. No. 151 to determine compliance with 3.3.

4.4.2 *Bending moment test.* Each sample unit shall be tested in accordance with the bending moment test of GGG-S-121 to determine compliance with 3.4.

4.4.3 *Blade-end toughness.* Each sample unit shall be subjected to the following test:

The flat tip ends of the blades shall be inserted in a vise to a depth equal to four times the tip thickness of the blade, and the brake tool pulled slowly as a lever until either a definite permanent bend or a fracture at the blade end results. If fracture occurs, the pieces shall be refitted and the blade end shall show that a definite noticeable permanent bend occurred prior to fracture. If no permanent bend of a fractured blade end is apparent, the blade end is brittle and the tool shall be considered to have failed the toughness test requirements.

4.5 **Performance test.**

4.5.1 *Size 1 brake tool.* Each sample unit of size 1 brake tools shall be tested on at least three varieties of automobiles with limited brake adjusting areas. The tool shall meet the requirements of 3.7.

4.5.2 *Size 2 brake tool.* Each sample of size 2 brake tools shall be tested on at least three varieties of automobiles with large size brake adjusting area. The tool shall meet the requirements of 3.8.

4.6 **Inspection of preparation for delivery requirements.** An inspection shall be made to determine that the preservation, packing, packaging, and marking comply with the requirements in section 5. Defects shall be scored in accordance with table II. For examination of interior packaging, the sample unit shall be one shipping container fully prepared for delivery, selected at random just prior to the closing operations. Defects of closure listed shall be examined on shipping containers fully prepared for delivery.

TABLE II. *Classification of preparation for delivery defects*

Examine	Defects
Markings (exterior and interior)	Omitted; incorrect; illegible; improper size, location, sequence, or method of application.
Materials	Any component missing or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling. Distortion of container.

5. PREPARATION FOR DELIVERY

(For civil agency procurement, the definitions and application of levels of packaging and packing shall be in accordance with Fed. Std. No. 102.)

5.1 **Preservation, packaging, and packing.** Unless otherwise specified (see 6.2), the tool shall be preserved, packaged, and packed in accordance with MIL-H-15424. The level of preservation and packaging shall be A or C and the level of packing A, B, or C as specified (see 6.2).

5.1.1 As part of the requirements of section 5 of this specification, the following is considered as part of table I of MIL-H-15424.

Column 1	Column 2	Column 3	Column 4	Column 5	Column 6
Tool, adjusting, brake shoe (spoon type)	P-1	I	1	36	

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5.2 Marking.

5.2.1 *Civil agencies.* In addition to markings required by the contract or order, the interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.2.2 *Military activities.* In addition to markings required by the contract or order, the interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The brake adjusting tools specified herein are of the spoon type for adjusting star nut brakes. Size 1 tools are intended for use on vehicles equipped with small, or limited, brake adjusting areas. Size 2 tools are intended for use on vehicles where space is not limited for making adjustment.

6.2 Ordering data. Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Size required (see 1.2.1, 3.7 and 3.8).
(c) Special configuration, if required (see 3.6 and figure 1).
(d) Levels of preservation, packaging, and packing required (see 5.1).

6.3 Transportation description. Transportation description and minimum weights applicable to this commodity are:

Rail:

Tools, not otherwise indexed by name.
Carload minimum weight 30,000 pounds.

Motor:

Tools, not otherwise indexed.
Truckload minimum weight 30,000 pounds, subject to Rule 115, National Motor Freight Classification

MILITARY INTEREST:

	Custodians	*Review activities*	*User activities*
Army	GL	GL	
Navy	YD		YD, MC
Air Force	69	69	

Preparing activity: GSA-FSS

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